

SLTC REASEARCH UNIVERSITY

SEMESTER 05

Bachelor of Applied Information Technology – (BAIT)

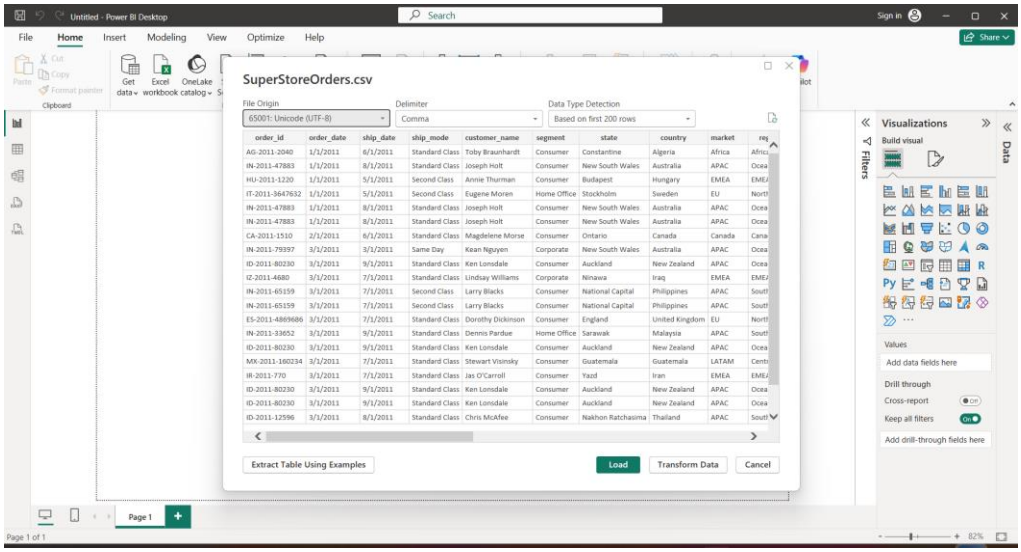
CIT308 - Data Visualization

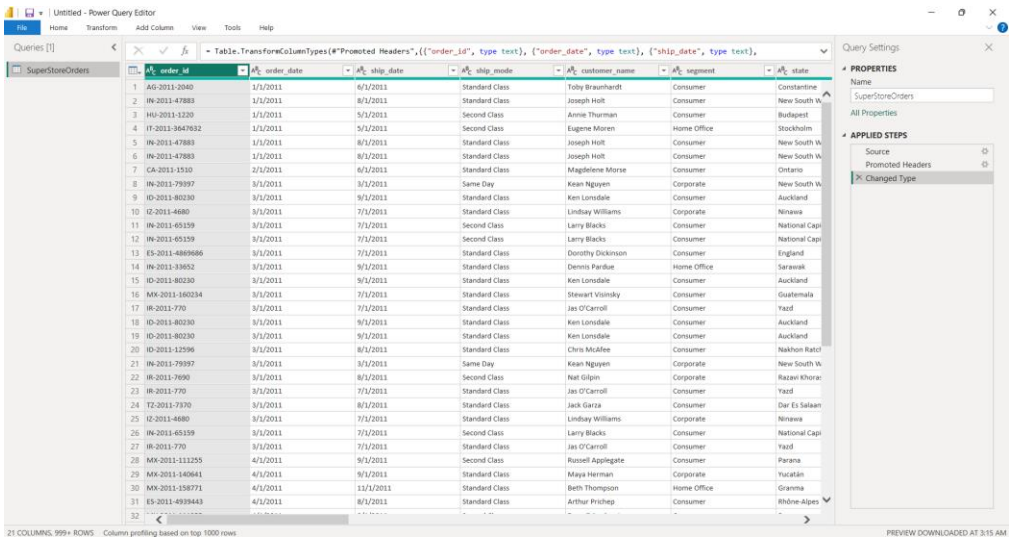
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Part 1 – Import Data

The Superstore Sales dataset (SuperStoreOrders.csv, 51,290+ records) was imported into Power BI Desktop using Get Data → Text/CSV. During import, column data types were verified and corrected (e.g. order_date and ship_date set to Date, sales/profit/discount/quantity/shipping_cost set to Decimal/Whole Number). The table was checked for duplicate rows and missing values using Power Query, and columns were renamed for clarity (e.g. 'Days to Ship' calculated column added).

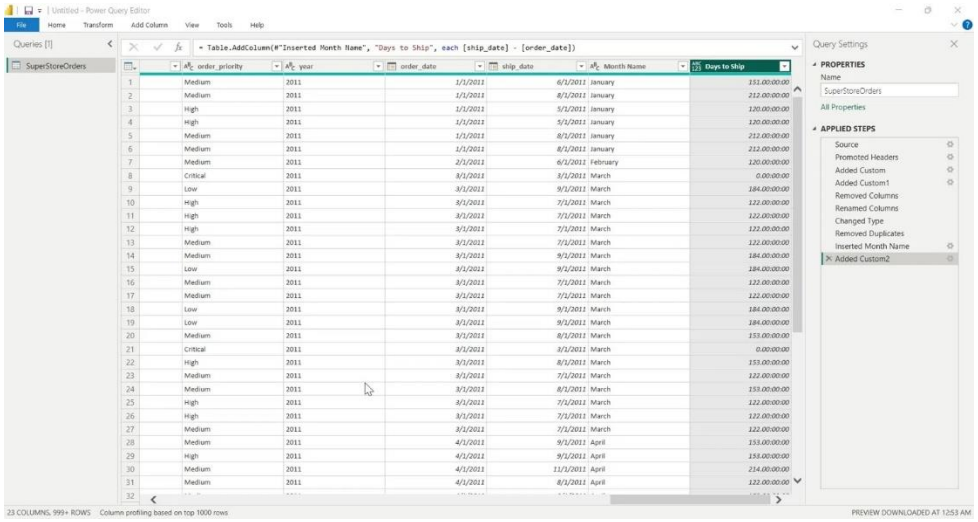


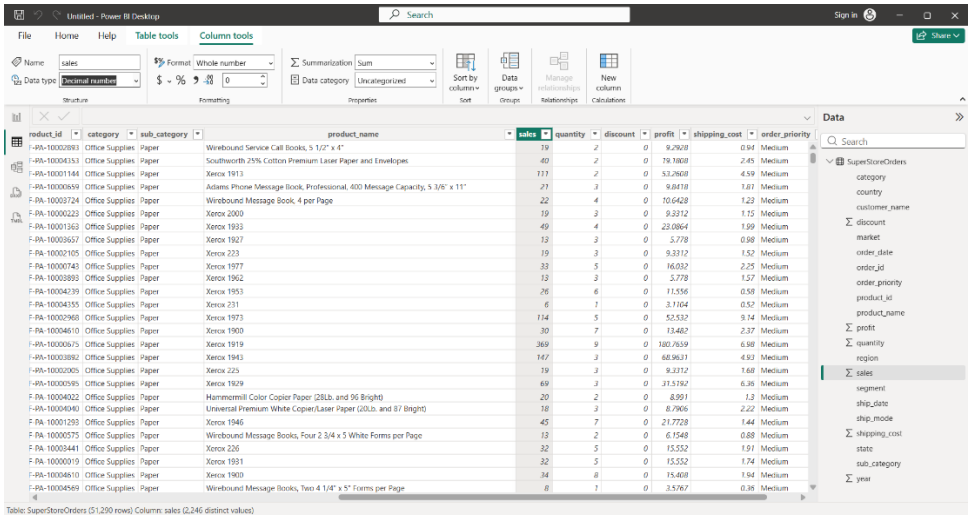


Part 2 – Data Cleaning

The following transformations were performed in Power Query:

- Removed duplicate rows using Remove Duplicates on order_id.
- Changed data types for sales, profit, discount, quantity and shipping_cost from Text to Decimal/Whole Number to fix calculation errors.
- Replaced null/blank values in relevant columns.
- Created a 'Month Name' column and a 'Days to Ship' calculated column (ship_date − order_date).
- Created a separate Date dimension table (DataTable) using CALENDAR(), with Year, Quarter, Month and Date columns extracted for time-based analysis.





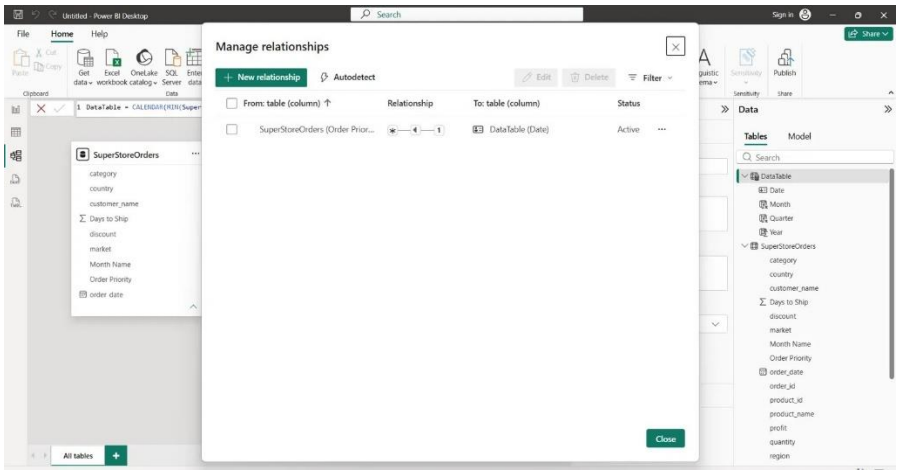
Part 3 – Data Modelling

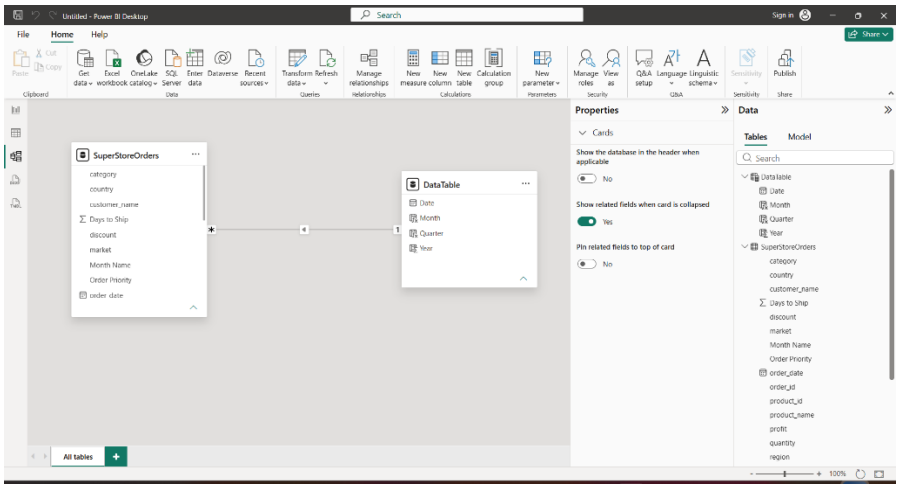
A separate Date dimension table (DataTable) was created using DAX and marked as a Date Table. It was related to the SuperStoreOrders fact table via the order_date column, using a one-to-many relationship (DataTable[Date] → SuperStoreOrders[order_date]), with DataTable as the 'one' side and SuperStoreOrders as the 'many' side and a single (one-directional) cross-filter direction.

Keys identified

- order_id — Primary key of each transaction in SuperStoreOrders.
- order_date — Foreign key linking SuperStoreOrders to DataTable[Date].
- Date — Primary key of the DataTable date dimension.

This model enables hierarchical drill-down (Year → Quarter → Month → Date) and time-intelligence DAX calculations.



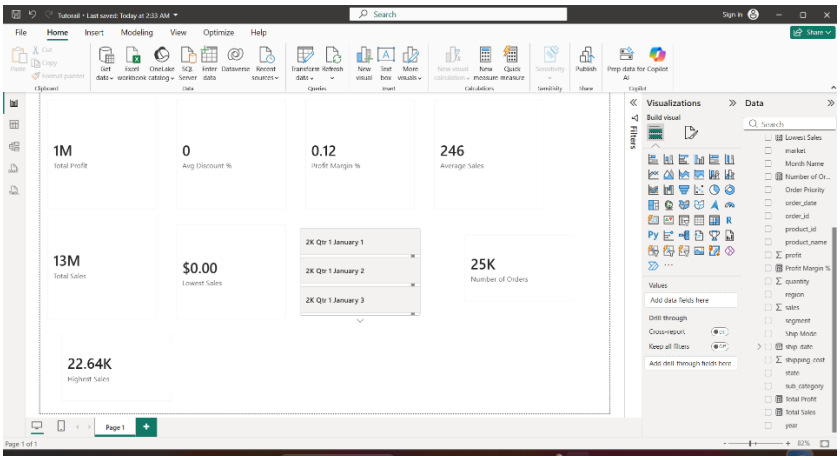


Part 4 – DAX Measures

The following measures were created on the SuperStoreOrders table:

- Total Sales = SUM(SuperStoreOrders[sales])
- Total Profit = SUM(SuperStoreOrders[profit])

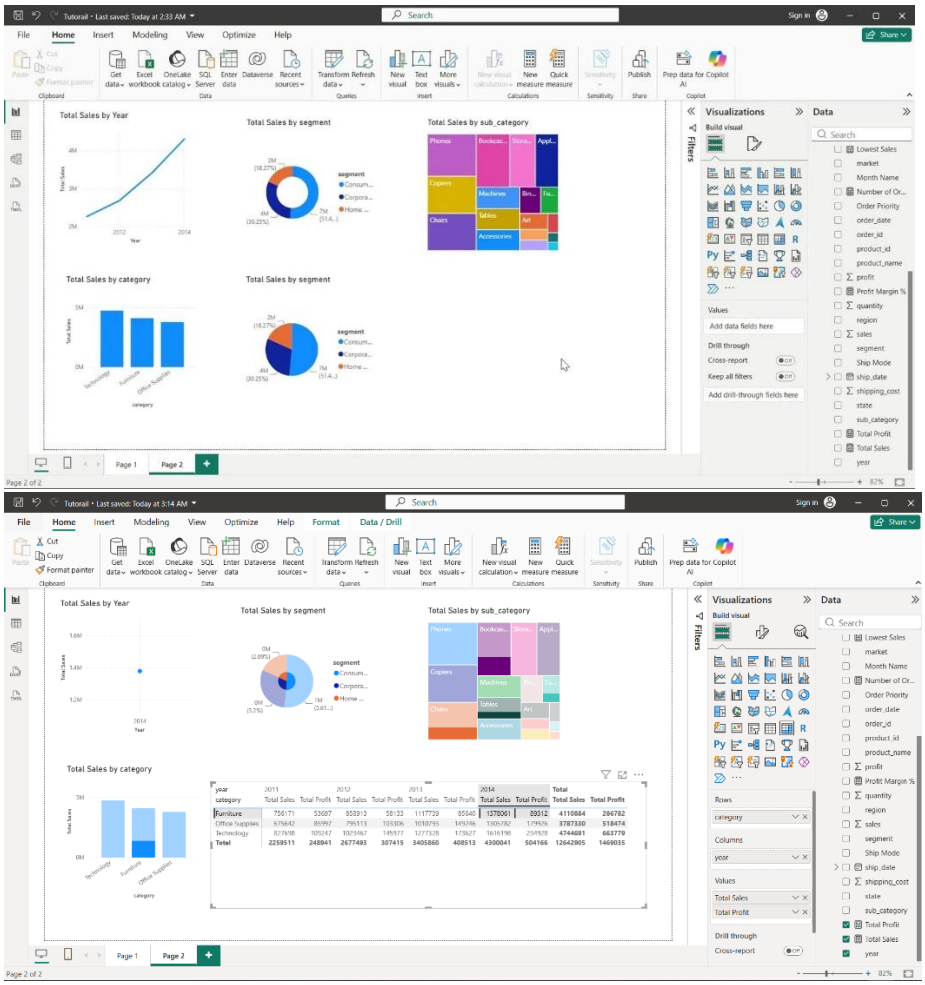
- Average Sales = AVERAGE(SuperStoreOrders[sales])
- Number of Orders = DISTINCTCOUNT(SuperStoreOrders[order_id])
- Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
- Highest Sales = MAX(SuperStoreOrders[sales])
- Lowest Sales = MIN(SuperStoreOrders[sales])
- Avg Discount % (custom KPI) = AVERAGE(SuperStoreOrders[discount])



Part 5 – Visualizations

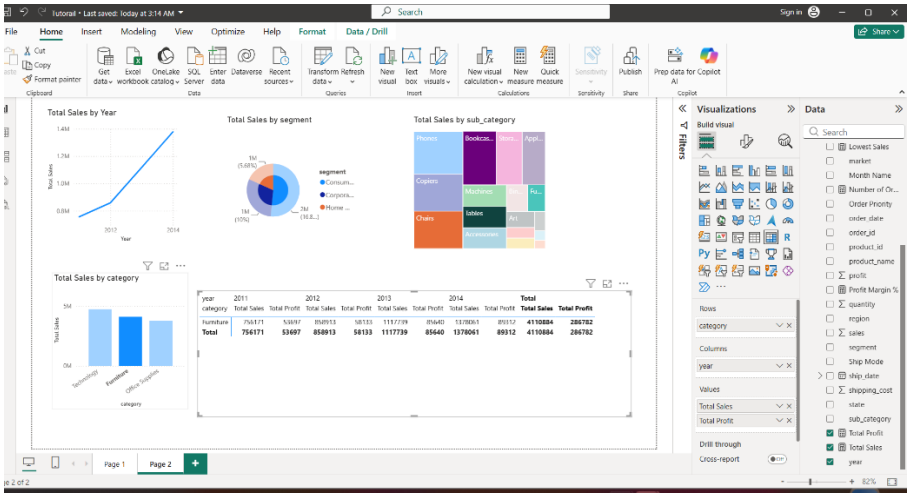
The following visuals were built on the report canvas:

- KPI Cards — Total Sales, Total Profit, Profit Margin %, Number of Orders
- Bar Chart — Total Sales by Category
- Line Chart — Total Sales by Year (trend 2011–2014)
- Donut Chart — Total Sales by Segment
- Treemap — Total Sales by Sub-Category



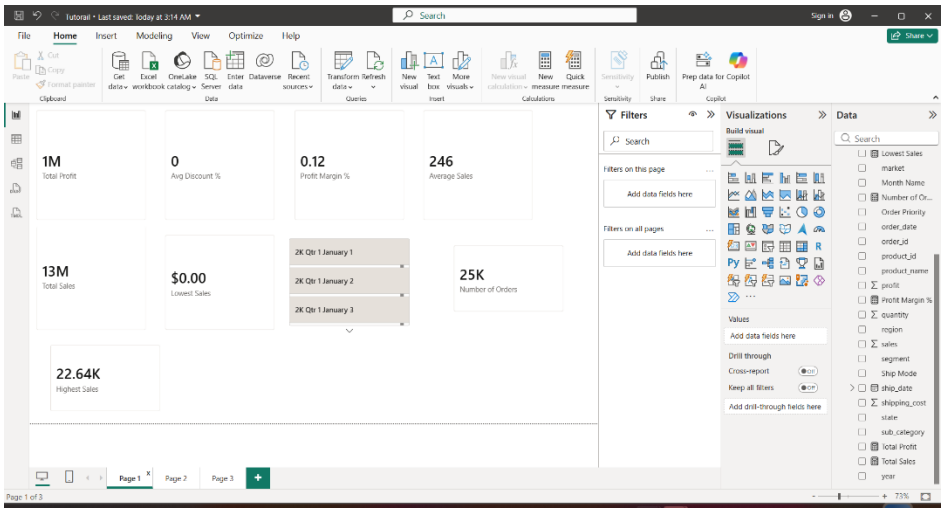
Part 6 – Interactive Dashboard

- Cross Filtering — Clicking any visual (e.g. a category bar) filters all other visuals on the page automatically.
- Tooltips — Default Power BI tooltips show exact values (Total Sales, Total Profit) on hover for each data point.
- Drill Down — The Year field on the line/matrix visuals supports drilling down to Quarter → Month using the Date dimension hierarchy.
- Slicer Interaction — The Year slicer filters all visuals on the page simultaneously.



Part 7 – Dashboard Design

A consistent Power BI theme was applied for uniform colours and fonts across all visuals. Titles were added to every visual, KPI cards were aligned in a single row at the top, and charts were arranged with consistent spacing/whitespace for a clean, single-page, professional dashboard layout.



Part 8 – Business Insights

- Sales show a consistent upward trend from 2011 to 2014, indicating steady business growth over the period.
- The Consumer segment contributes the largest share of total sales (around 51%), making it the most valuable customer segment.
- Technology and Furniture categories generate strong sales, while Office Supplies has high order volume but comparatively lower profit per order.
- A small number of sub-categories (e.g. Phones, Chairs, Copiers) account for a disproportionately large share of total revenue, as seen in the treemap.
- The scatter plot of Profit vs Sales shows that higher sales do not always translate to proportionally higher profit, suggesting some products are sold at high discounts or low margins.